

# SHLOK THAKKAR

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## EDUCATION

### University of Illinois Urbana-Champaign

B.S. in Computer Science + Economics, GPA: 3.91/4.00, Minor in Statistics

Expected May 2028

Champaign, IL

**Awards:** 2nd Place, HackIllinois 2026 | 2nd Place, HERE Chicago Hackathon 2026 | Finalist, Builders Cup by Ramp 2026

## EXPERIENCE

### Johnson Controls, AI Lab

Sep. 2026 – Present

Software Engineering Intern

Champaign, IL

- Increased RAG answer accuracy from 68% to 79% for LLM agents over building-automation data by implementing hybrid search and reranking across equipment telemetry and technical manuals
- Built Python LLM agent services that unify time-series telemetry and PDF manuals into one cited-answer query interface

### Stealth Startup (Product Hunt)

Jun. 2026 – Aug. 2026

Software Engineering Intern

New York, NY

- Launched a consumer AI claims web application to #5 Product of the Day on Product Hunt (231 upvotes) as one of three engineers, reaching 350+ users and recovering \$20,000 for them while holding a 0.032% AWS Lambda error rate in production
- Architected a scalable serverless REST API across 4 AWS Lambda functions and Aurora PostgreSQL Serverless v2 provisioned with Terraform, sustaining sub-100ms query latency while removing 4 managed services via the RDS Data API
- Engineered a reliable 8-stage claim-lifecycle state machine on Inngest with typed events, per-case concurrency isolation, and automatic retries, covered by 57 unit tests
- Designed a 12-table PostgreSQL schema with pgvector (1,024-dim embeddings) and full-text search, powering hybrid retrieval via RRF fusion of cosine similarity and BM25 in a single SQL query with no separate vector store

### Disruption Lab @ UIUC

Aug. 2025 – Present

Tech Lead (previously Software Engineer)

Champaign, IL

- Directed two 12-week AI course-intelligence projects for faculty clients as Tech Lead, owning design through delivery of a Neo4j prerequisite knowledge graph with RAG roadmap generation and a Canvas API skill-extraction pipeline
- Redesigning a read-only research tool into a fully editable platform for 40+ users by shipping 15+ REST API endpoints over a Neo4j graph, cutting API latency 60% (300ms to 120ms) with optimized Cypher queries
- Expanded automated tax-schema diff coverage from 7 to 27 states for a Fortune 500 financial-services client with a Python and lxml XSD parser, validated by 43 unit tests

### Parallel Programming Laboratory, UIUC

Apr. 2026 – Present

Undergraduate Research Assistant

Champaign, IL

- Developed custom operators and broadcasting in C++20 for a Charm++ distributed NumPy array abstraction across multi-node clusters
- Redesigning core runtime APIs for Reconverse, a ground-up C++ rewrite of the Charm++ messaging and scheduling layer

## PROJECTS

### raft-kv | Go, net/rpc, GitHub Actions

GitHub

- Implemented Raft consensus from scratch in Go over an RPC network layer (leader election, log replication, persistence, snapshotting) behind a 5-node distributed key-value store with linearizable reads and exactly-once client semantics
- Verified fault tolerance under 1,000+ concurrent operations with network-partition and chaos tests that kill and rejoin nodes across 30-second fault windows, run under the Go race detector in CI/CD

### Limit Order Book | C++20, CMake, Catch2

GitHub

- Designed a thread-safe C++20 matching engine across 5 order types with FIFO priority, achieving O(1) cancellation and an incremental level cache that cuts pre-checks from O(N) to O(P) (roughly 500x on a 10K-order book)
- Sustained low-latency reads with O(1) microstructure queries (best bid/ask, spread, mid-price, depth), validated by 50 Catch2 test cases

### vorssaint/vorssaint-utils (Open Source) | Swift, macOS

Merged PRs

- Merged 2 Swift fixes into a macOS toolkit with 19K+ GitHub stars (steady window-switcher icon rows, window fronting via the window server) and opened 3 more PRs (hidden-app placement, Spotify cache handling, Homebrew proxy settings)

## TECHNICAL SKILLS

**Languages:** C/C++ (C++20), Java, Python, Go, TypeScript, JavaScript, C#, SQL, Swift

**Systems & Networking:** Distributed Systems, RPC, Fault Tolerance, Raft, Charm++, Multithreading, Object-Oriented Design

**Database Systems & Data:** PostgreSQL (SQL), pgvector, Neo4j (NoSQL), Supabase, Query Optimization, Data Pipelines, FastAPI

**AWS & Operations:** Amazon Web Services (AWS: Lambda, Aurora, S3), Terraform, Docker, Kubernetes, CI/CD (GitHub Actions)

**AI-Assisted Development:** AI Coding Tools (Claude Code), LLM Agents, Retrieval-Augmented Generation (RAG), LangChain

**Relevant Coursework:** Data Structures, Algorithms & Models of Computation, Computer Architecture, System Programming